

Resultant forces, work done and energy transfer

Forces
Paper 2 · Combined and Separate
16 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

	QUESTION	ANSWER	✓
1	Define resultant force.	The single force that has the same effect as all the forces acting on an object.	☐
2	Describe how a resultant force is calculated.	By adding all the forces on the object as vectors.	☐
3	Two forces act in the same direction. Describe how to find the size of the resultant force.	Add the two forces together.	☐
4	Two unequal forces act in opposite directions. State the direction of the resultant force.	The direction of the larger force	☐
5	Two forces act in opposite directions. Describe how to find the size of the resultant force.	Subtract the smaller force from the larger force.	☐
6	State the resultant force when the forces on an object are balanced.	Zero (0 N)	☐
7	State the SI unit of work done.	Joule (J)	☐
8	State the SI unit of force.	Newton (N)	☐
9	State the SI unit of distance.	Metre (m)	☐
10	State a unit that is equivalent to 1 J of work done.	1 N m	☐
11	Work is done against friction. Name the energy store that increases.	The thermal energy store	☐
12	Describe the effect on an object's temperature of doing work against friction.	Its temperature rises.	☐

QUESTION		ANSWER	✓
13	State the equation linking work done, force and distance.	work done = force $\times$ distance (along the line of action of the force) ($W = F s$)	☐
14	State how much work is done when a force of 1 N moves an object through 1 m.	1 J	☐
15	Describe what a free body diagram shows.	All the forces acting on one object, drawn as arrows.	☐
16	State what is meant by resolving a force.	Splitting it into two components at right angles that together have the same effect.	☐